Evolutionary Computation and its Applications

Today's goals

Module 1
Fundamental Concepts in Optimization

Module 2
Genetic Algorithm

Module 5 Introduction

Introduction to Module 5's Study
Video • 1 min

How to Access MATLAB Online
Reading • 5 min

Module 5-1 Genetic Algorithm 1

Genetic Algorithm: A First Introduction
Video • 6 min

Genetic Algorithm: A Demo
Video • 8 min

Materials on Genetic Algorithms
Reading • 10 min

How Genetic Algorithms Solve Optimization Problems
Reading • 5 min

Module 5-2 Genetic Algorithm 2

Genetic Algorithms: GA Representation, Initialization and Selection
Video • 10 min

Genetic Algorithms: GA Crossover and Mutation
Video • 7 min

Genetic Algorithms: GA Stopping Criterion and Summary
Video • 5 min

Materials on Genetic Algorithms
Reading • 10 min

How GA Operators Work Together
Reading • 5 min

Module 5-3 Use Genetic Algorithm 1

Use GA via MATLAB: GA Function
Video • 7 min

Use GA via MATLAB: Employ the Function
Video • 6 min

Materials on Using Genetic Algorithms
Reading • 10 min

Using Genetic Algorithms Effectively in MATLAB
Reading • 5 min

Module 5-4 Use Genetic Algorithm 2

Use GA via MATLAB: Case Studies
Video • 5 min

Materials on Case Studies
Reading • 10 min

Solving Real Engineering Problems with Genetic Algorithms
Reading • 5 min

Module 5 Quiz

Module 5 Quiz
Practice Assignment • 30 min

Module 5 Assignment

Module 5 Assignment
Graded App Item • 30 min

Objective function for Assignment 5
Reading • 5 min

Applying Genetic Algorithms to a Real Engineering Design Problem
Reading • 5 min

Designing and Applying a Genetic Algorithm to Solve an Engineering Optimization Problem
Dialogue • 5 min

Module 5 Recap
Reading • 5 min

Module 3
Particle Swarm Optimization

Using Genetic Algorithms Effectively in MATLAB

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In this section, you learned how to implement genetic algorithms in MATLAB using the ga() function and explored how different inputs and options influence optimization behavior and results.

Reflect on the following questions:

- Why is it important to explicitly define **bounds** instead of relying on MATLAB's default search ranges?
- How does converting a maximization problem into a minimization problem affect the way you define the objective function in ga()?
- In the Rastrigin example, why did adding bounds and options significantly improve the quality of the solution?
- How does enforcing **integer constraints** using intcon change the nature of the optimization problem and the search process?

This discussion will help you move beyond simply running ga() and toward using it thoughtfully and effectively in real engineering and scientific applications.

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